PLM - Engineering Changes, Executed with Control

1. Overview

PLM is an Engineering Change Order (ECO) system that enables **controlled, versioned, and approval-driven changes** to Products and Bills of Materials.

It prevents direct edits to master data by ensuring all changes are **proposed, reviewed, approved, and applied through ECOs**, keeping active data stable, preserving history, and allowing only approved changes in operations.

2. Key Challenges in Existing Systems

- Direct Product and BoM edits without version control
- No separation between draft and applied changes
- Loss of traceability due to overwritten data
- Missing or inconsistent approval workflows

3. User Roles & Access Control

Role	Access
Engineering User	Create and modify ECOs,Proposes Product or BoM changes, Works on draft versions, Initiates approval workflows
Approver	Reviews proposed changes, Approve / validate ECOs based on stage rules, Controls when changes become effective
Operations User	View active Products & BoMs only
Admin	Full access, Configures ECO stages and approval rules, Manages master data settings

4. Navigation & System Entry

Authentication

- Common login and signup flow
- Users land on the PLM dashboard after login
- UI behavior adapts based on role and permissions

Master Menu (Left Sidebar)

- Products
- Bills of Materials
- Engineering Change Orders (ECO)
- Reports

- Settings: ECO Stages, Approval Rules

5. Core Modules & Functionalities

Product Master: Stores the base definition of a product.

Fields: Product Name, Sale Price, Cost Price, Attachments, Current Version (automated), Status (Active / Archived)

Behavior

- Products are version-specific
- Archived products:
  - Are read-only
  - Cannot be selected in BoMs or ECOs
  - Remain visible for traceability

Bill of Materials (BoM): Represents the manufacturing structure of a product version.

Fields: Product, BOM Version, Components (Product + Quantity), Operations (Time, Work Center), Status (Active / Archived)

Behavior

- BoMs are version-specific
- Only BoMs linked to Active product versions are usable
- Old BoMs remain preserved after version updates

Engineering Change Order (ECO): Central object used to manage and apply changes.

Fields: Title, ECO Type (Product / BoM), Product, Bill of Materials, User, Effective Date, Version Update (checkbox), Stage

Behavior

- ECOs start in a **New** stage
- Mandatory fields must be filled before starting
- Version Update:
  - Enabled → Creates a new version
  - Disabled → Applies changes to the same version
- Changes are stored separately and do **not** affect master data until approved

ECO Stages & Approvals: Stages are configurable via Settings.

Examples: New, Approval, Done
Rules:

- Approval requirements are stage-based
- If approval is mandatory:
  - Approval button is shown
- If no approval is configured:
  - A **Validate** button is shown instead

- Approving an ECO:
  - Moves it to the next stage
  - Final stage marks ECO as applied

Versioned Change Comparison: Before approval, users can review changes using **diff-style views.**

BoM Change Comparison

- Component-wise quantity comparison
- Old vs New version
- Visual indicators:
  - Green → Added
  - Red → Reduced
  - Neutral → Unchanged
- Operation time changes shown clearly

Product Change Comparison

- Sale Price
- Cost Price
- Attachments
- Side-by-side version comparison

6. Lifecycle Enforcement Rules

- Draft or under-approval data cannot be used in operations
- Only:
  - Active Product
  - Active Version
  - Active BoM

  can be used downstream
- Once an ECO reaches the final stage:
  - New version becomes Active
  - Old version is Archived
- Archived data is immutable and retained for audit

7. Audit & Traceability: Every critical action is logged.

- **Tracked Events:** ECO creation, Stage transitions, Approval actions, Version creation, BoM changes
- **Audit Fields:** Action, Affected record, Old value → New value, User, Timestamp

8. Reporting

Engineering Change Orders Report

Columns: ECO Title, ECO Type, Product, Changes (clickable)
Clicking **Changes** opens the detailed comparison view for that ECO.

Additional Reports

- Product Version History
- BoM Change History
- Archived Products
- Active Product–Version–BoM Matrix

9. Example Scenarios

Scenario 1: BoM Quantity Adjustment

Product: iPhone 17
BoM Version: v1
ECO raised to reduce component quantity for cost optimization.

Result:

- New BoM Version created
- Old version archived
- Product remains the same
- Changes visible in comparison view

Scenario 2: Product Pricing Update

Product: iPhone 17
ECO Type: Product
Sale price and cost price updated.
Result:

- New product version created
- Old version archived
- Operations continue using only the new version

10. Outcome: PLM ensures:

- Controlled and auditable changes
- No accidental overwrites of master data
- Clear separation between **proposed**, **approved**, and **applied** changes
- High confidence in manufacturing and operational data

11. Key PLM Concepts Mapping

Manufacturing Concept	PLM Object
Product Master	Product
Bill of Materials	BoM
Work Orders	Operations / Routings
Change Request	ECO

Approval Flow	ECO Stages
Change History	ECO Logs

12. Applying ECOs on Bill of Materials (BoM)

When to Use a BoM ECO: Use a BoM ECO when:

- Component quantity changes
- New raw material is added
- Component is removed
- Operations are modified

Example Scenario – Wooden Table BoM Change

Existing BoM (Version 1)

Product: Wooden Table
Quantity: 1 Unit

Components

- 4 × Wooden Legs
- 1 × Wooden Top
- 12 × Screws
- 1 × Varnish Bottle

Operations

- Assembly – 60 mins @ Assembly Line
- Painting – 30 mins @ Paint Floor
- Packing – 20 mins @ Packaging Line

13. Creating an ECO for BoM Change

Step 1: Create ECO

- ECO Type: BoM Change
- Product: Wooden Table
- BoM Version: Current

Step 2: Propose Changes

- Change Screws from **12 → 16**
- Add **Quality Inspection – 10 mins**

Step 3: Review & Approval

- Engineering reviews feasibility
- Manufacturing reviews capacity
- Manager approves ECO

Step 4: Apply ECO

- New BoM Version is created
- Old BoM is archived
- Future MOs use updated BoM

Existing MOs remain unaffected (Just a heads-up.. not part of the problem statement)

14. Applying ECOs on Products

When to Use Product ECO: Use Product ECO for:

- Product name change
- Price update
- Attachment Update

Example – Product Change ECO

Scenario

- Wooden Table now uses Premium Finish
- Cost increased due to varnish upgrade

ECO Changes

- Product Cost: Updated

After approval:

- Product master is updated
- Stock valuation rules apply
- Future MOs use updated product data

15. Why is this Hackathon Problem Important?

- Exposes students to real-world PLM and ECO workflows.
- Demonstrates product & BoM versioning with approvals, not direct edits.
- Shows how changes flow safely across **Engineering** → **Manufacturing** → **Operations**.
- Emphasizes business logic, traceability, and lifecycle control over simple coding.

Mockup: https://link.excalidraw.com/l/65VNwvy7c4X/5DpaV3CG8zv